

L4

Motor Movement

1. The *degrees of freedom*: How are movements selected out of infinite options to achieve particular tasks?
2. The *sequencing and timing* : How are behaviors ordered in time?
3. The *perceptual-motor integration* : How are perception and motor control combined?
4. The *learning* : How are perceptual-motor skills acquired?

THE DEGREES OF FREEDOM PROBLEM

The degrees of freedom in a system are the ways the system can independently vary.

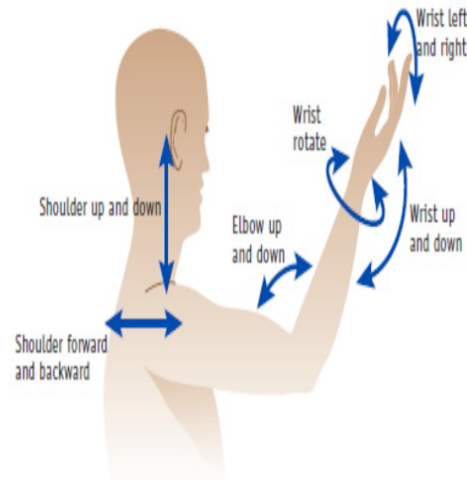
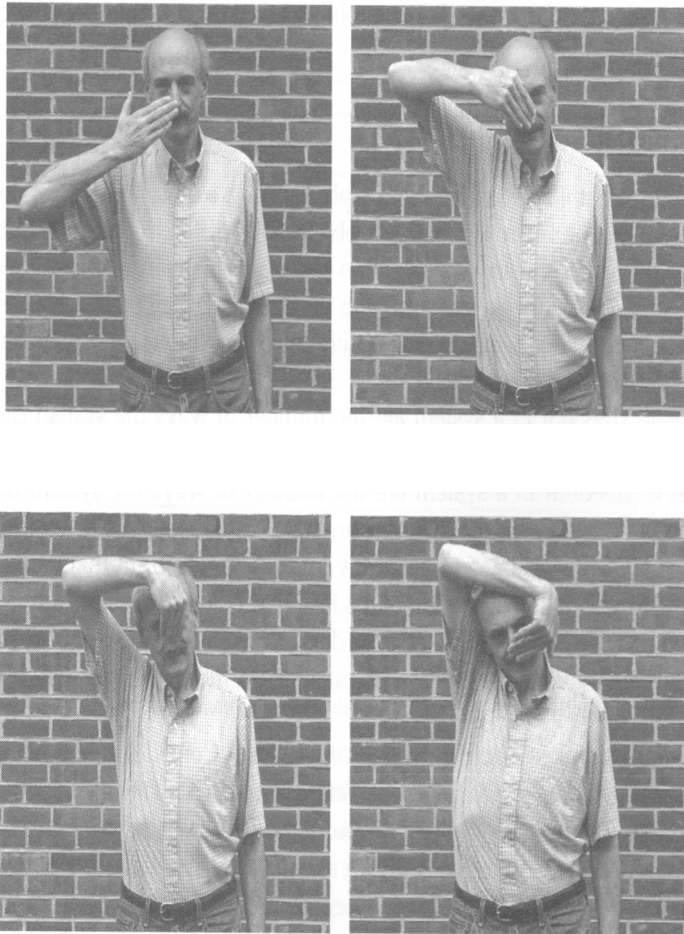
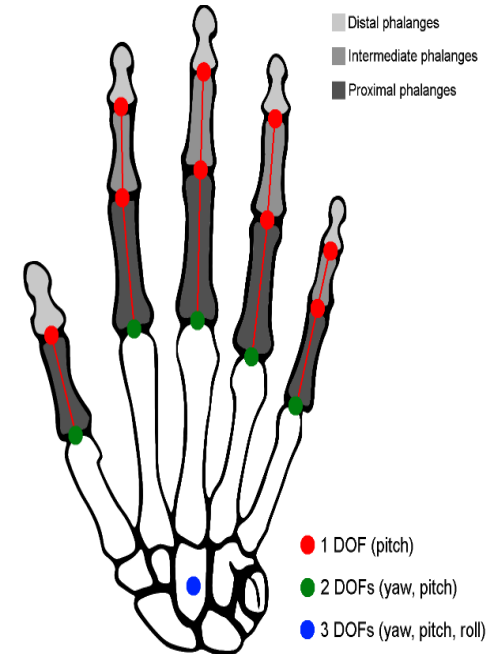


Figure 14.24 Degrees of freedom in the human arm

<http://web1.muirfield-h.schools.nsw.edu.au/mahara/view/view.php?id=14089>



Moreira et al, 2015

- ***kinematics*** (positions in time)
- ***Kinetics*** (forces in time)
- **Mapping of neural activity to muscle activity.**

Many-to-one mapping

Synergies

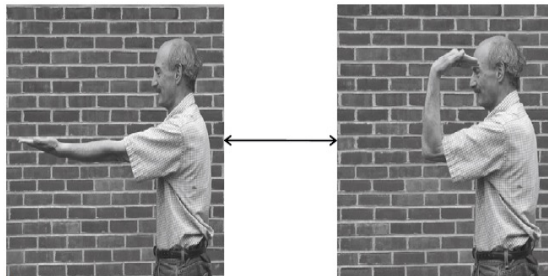
Coordination *within* limbs

Behaviour of one joint influences other

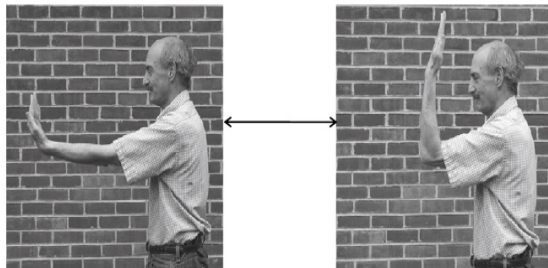
Functional linkages between different parts of the body



<https://gfycat.com/equalbeneficialallosaurus>



A



Synergy $\longleftrightarrow$ **DOF**

Degree Of Freedom

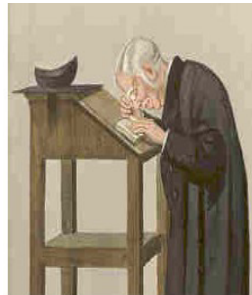
- **Synergy**
- **Mechanics**
- **Efficiency**

THE SEQUENCING AND TIMING PROBLEM

Control of sequencing and timing of our behaviour



Spooner and Student



William A. Spooner

"**roaring pain**" **instead of** "**pouring rain**."

A **spoonerism** is an error in speech in which corresponding consonants, vowels, or morphemes are switched between two words in a phrase.

Action Slips: wrong motor activity

When we engage in behaviors that have distinct elements, such as speaking, typing, or walking, the elements of the behaviors must be ordered correctly in time . How is such ordering achieved?

SEQUENCING: Coarticulation & Timing

Simultaneous motions that occurs in sequential tasks



<https://www.online-tech-tips.com/computer-tips/best-methods-to-improve-your-typing-speed/>



<https://mindfulpresenter.com/7-takeaways-barack-obamas-better-presenter/>



<https://www.musicradar.com/how-to/improve-your-acoustic-folk-guitar-playing-in-20-minutes>

Effective way of increasing response speed

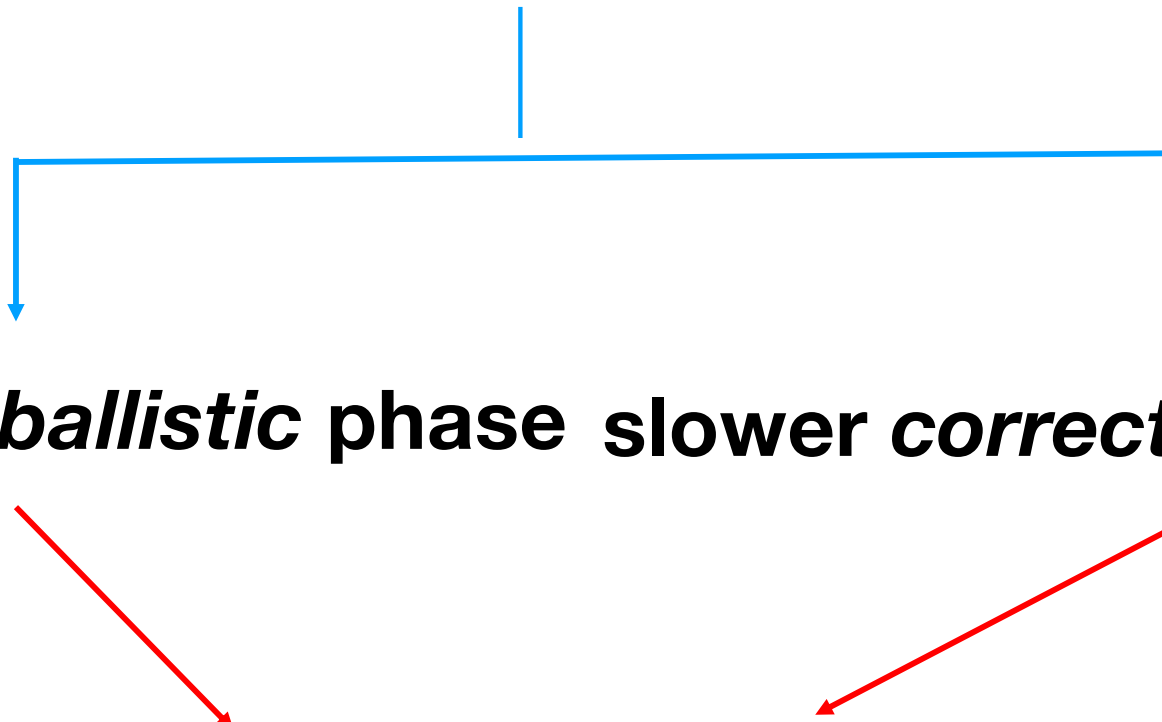
THE PERCEPTUAL-MOTOR INTEGRATION PROBLEM

Perception ↔ **Movement**

Movement

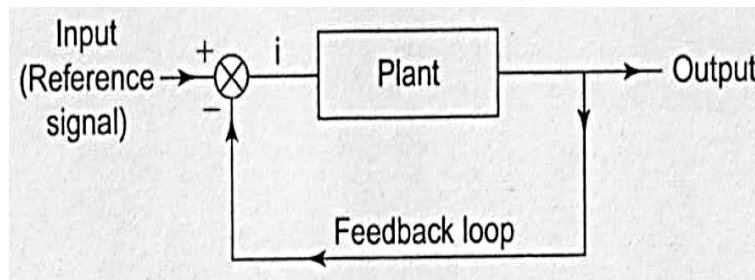
rapid *ballistic* phase **slower *corrective* phase**

Movement control

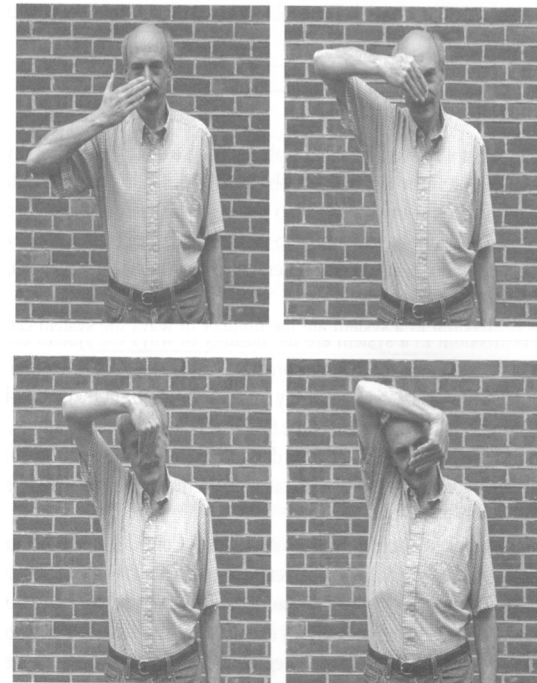


Servo mechanism

Correcting errors through negative feedback loop



- Close loop control
- Open loop control



Feedforward control

Movement executed without feedback

Have you ever seen your own eye movement on mirror?



saccadic suppression

Saccadic suppression

A saccade is a quick, simultaneous movement of both eyes between two or more phases of fixation in the same direction. (*Wikipedia*)



<https://en.wikipedia.org/wiki/Saccade>



Helps reducing the damage to visual perception caused by such retinal smearing.

Movement executed without feedback

Feedforward control

Helps distinguish
perceptual changes due to motion arising in the external
environment versus perceptual changes arising from one own's
motion.

Reafference: perceptual changes arising from one own's motion.

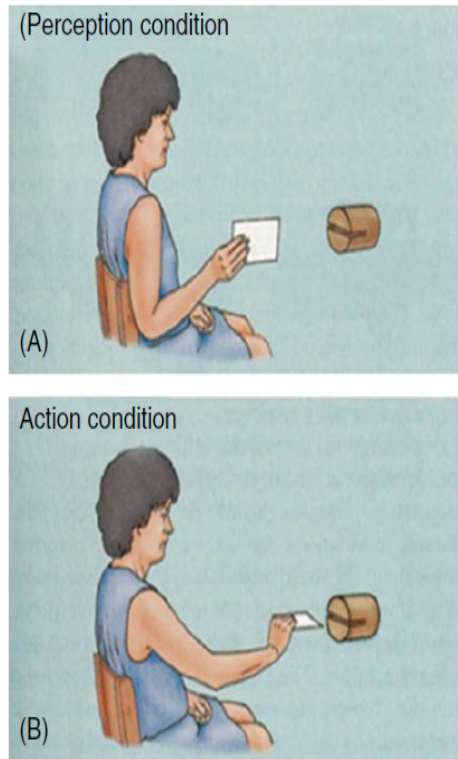
Exafference: perceptual changes due to motion arising in the external
environment

Movement & Perception

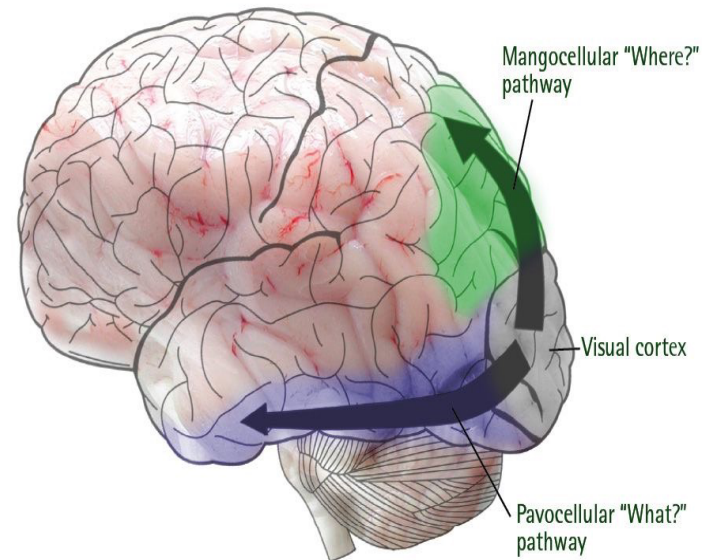
Perception  **Movement**

- **Movement enhances perception**
- **Movement informs perception**
- **Mirror neurons**

Movement enhances perception



Milner & Goodale, 1995



<https://aspergerhuman.wordpress.com/tag/visual-pathways/>

Perception ↔ Movement

Movement informs perception



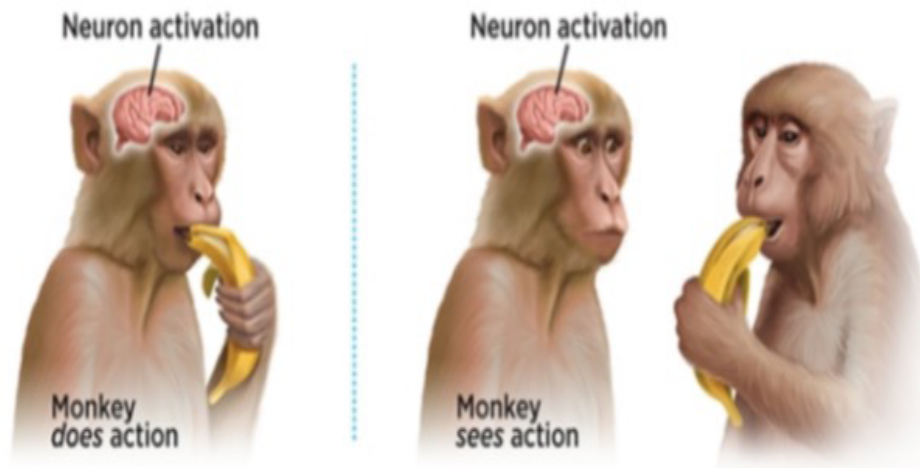
Proffitt et al, 2006

Perception $\longleftrightarrow$ **Movement**

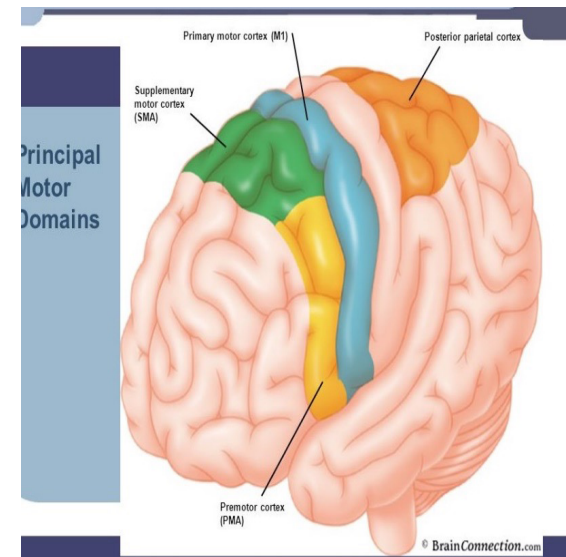
Steepness perceived more with weight than without weight.

Mirror Neurons

A mirror neuron is a neuron that fires both when an animal acts and when the animal observes the same action performed by another.

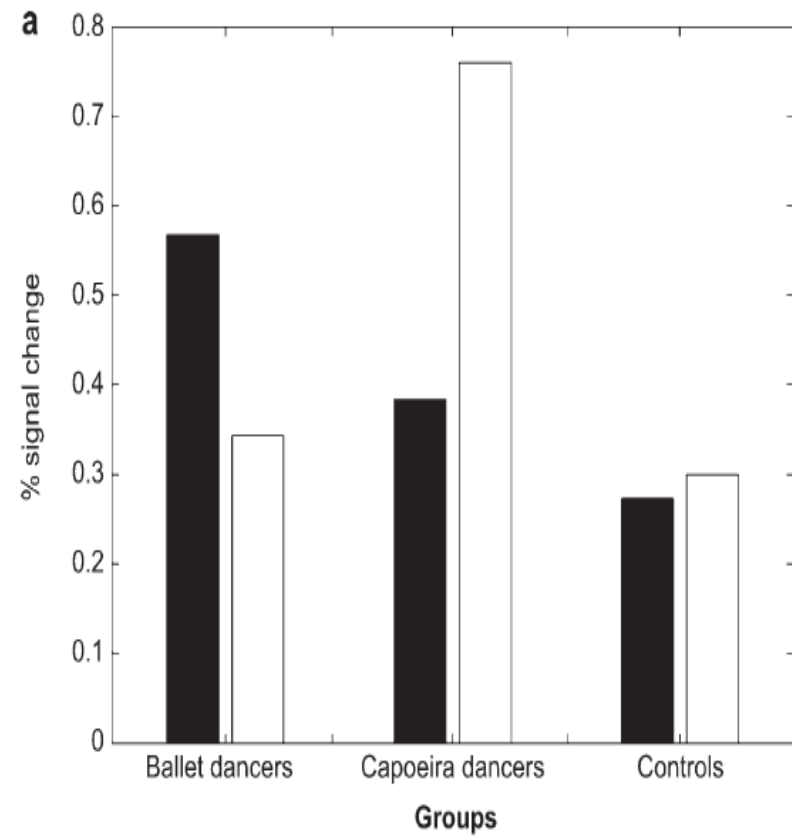
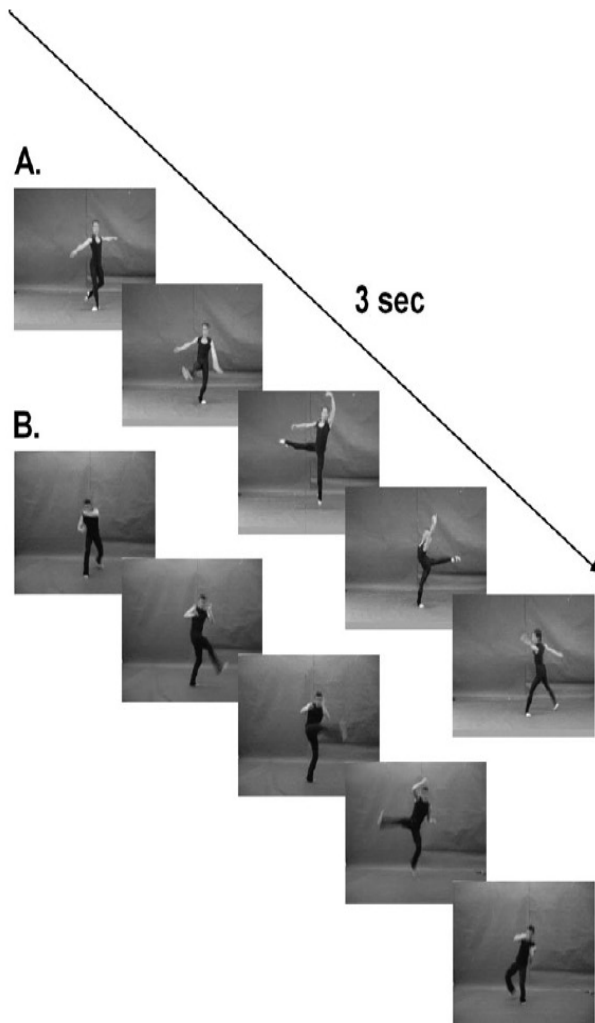


<https://critpoints.net/2016/08/03/mirror-neurons/>

[illegible]

- Learning
- Intention
- Empathy

Mirror Neurons



□ Capoeira stim
■ Ballet stim

Merino et al, 2005

Motor Movement

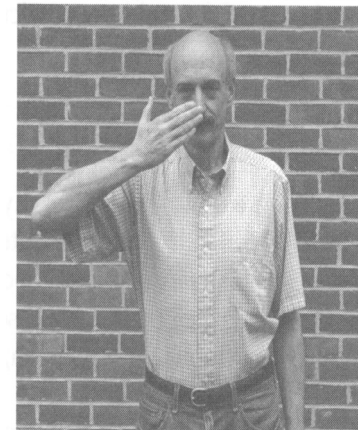
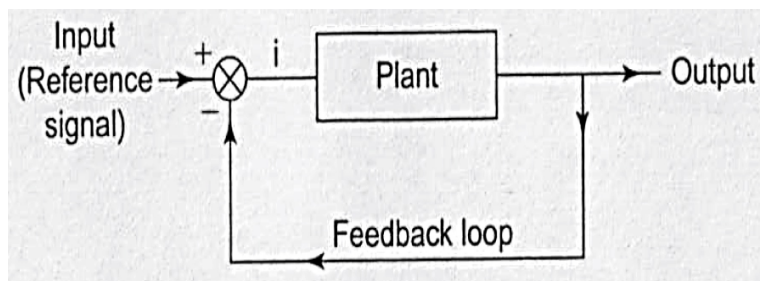
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Learning

How are motor skills acquired?

- By doing
- Practising deliberately
- Specificity
- Neural plasticity (economical use of brain)

Servo mechanism (Closed- loop)



**Learning effective with
feedback**